NAME

MBgrdviz − Simple interactive 2D/3D visualization of GMT grids.

VERSION

Version 5.0

SYNOPSIS

MBgrdviz
[--grid-file=grdfile {-Igrdfile}
--help {-H}
--overlay-file=grdfile {-Jgrdfile}
--test {-T}
--verbose {-V}]

DESCRIPTION

Overview

MBgrdviz is an interactive 2D/3D visualization tool for **GMT** topography grid files. **MBgrdviz** can be used for general data visualization and also has features integrating it with swath mapping data processing and swath survey planning.

Once a topographic grid has been read, it may be displayed either as a 2D map or as a 3D model. The 2D view can be panned and zoomed; the 3D view also allows arbitrary rotations of both the topography model and the viewpoint.

In addition to the primary topography, other data sets (e.g. sidescan sonar mosaics) can be read and displayed as overlays on the topography. Navigation of ships or underwater vehicles can be read and displayed; if associated with swath data the extent of mapping coverage can be displayed. Site data (individual point locations) can be read, displayed, interactively created and modified, and saved. Route data (collections of waypoints) can also be read, displayed, interactively created and modified, and saved. GIS polylines, polygons, and points can be read from MB-System files or imported from ESRI shapefiles, displayed, and saved or exported, but not edited.

The topography data can be shown using a number of colortables. The colortables can be linearly stretched between minimum and maximum values, or applied using a histogram equalization. The display may also be shaded using an illumination model, slope magnitude, or overlay grid data. Overlay data can be displayed draped on the topography. Topographic contours can be displayed using a user defined contour interval.

In addition to modifying the view, users can interact with the display in several ways. Users can pick single points on the surface to determine position and topography values, or pick and drag to determine range and bearing between two locations. Arbitrary rectangular regions with north-south and east-west boundaries or arbitrarily oriented rectangular areas can be defined by picking and dragging. The data within a region can be opened as a new display; areas can be used for survey planning.

Topographic profiles associated with two-point picks, selected navigation, or selected routes can be displayed in a separate window and saved.

MBgrdviz can be used to select swath data files for editing or analysis using other **MB-System** tools. If survey navigation has been read and displayed, a user may interactively select files of interest, and then start up the interactive waterfall bathymetry editor **MBedit** to work on those files. Also available are the interactive, visualization based bathymetry editor **MBeditviz**, the navigation editor **MBnavedit**, and the water sound speed modeling tool **MBvelocitytool**.

MBgrdviz supports as many as ten displays deriving either from reading topography grids or extracting user defined regions from existing displays.

Starting Up

When **MBgrdviz** is invoked on the command line, a single window comes up with the title **MBgrdviz** and two menus: **File** and **Help**. The **Help** menu contains a single item named **About**, which brings up a simple information window including the MB-System release version identifier. The **File** menu contains several items, but initially only the **Open Primary Grid** and **Quit** items are enabled. Selecting **Open Primary Grid** brings up a filesystem browser window that allows users to find and select the desired **GMT** topography grid. By default, the browser filter is set to display only files ending with a ".grd" suffix. Users can change this filter (e.g. setting it to "*" will display all files) and then hit the **Filter** button to relist the files as desired. Once the desired topography grid file has been located and selected so that its full path appears in the **Selection** text item, clicking the **OK** button will cause **MBgrdviz** to read the grid and open an **MBview** display window. Initially, the topography will be displayed in 2D, or map view mode. The default color table will be "Haxby", and the topography will be shaded using slope magnitude.

Picking, Zooming, Panning in 2D View

MBgrdviz will initially be in the "pan and zoom" mouse mode.

In this mode, if the cursor is over topography when the left button is clicked, a red X will appear at the pick point and an info window on the left side of the **MBview** display window will show the longitude, latitude, and topography value of the pick point. If the left button is used to drag the cursor from one location to another, both the pick and release points will be shown as red Xs and a red line will be displayed in between. The info window will show both the start and end positions and also the range and bearing between the two locations.

The middle button can be used to pan by clicking, holding down, and moving the cursor.

The right button can be used to zoom the display by clicking, holding down, and moving the cursor up to zoom in or down to zoom out.

Manipulating a 3D View

The user can switch from 2D map view to 3D view mode using the <View->3D Display> menu item (and can switch back using the<View->Map Display> menu item). In the 3D view mode, picking, panning, and zooming are done in the same fashion as in the map view. However, the display is a perspective view that includes both rotation of the topographic model and of the viewpoint. To rotate the model, select the "Rotate Model" button (or the <Mouse->Rotate Model> menu item). The left button can be used to pick topography, the middle button will rotate the model when clicked and dragged, and the right button will change the vertical exaggeration when clicked and moved up (more exaggeration) or down (less exaggeration). To rotate the viewpoint, select the "Rotate View" button (or the <Mouse->Rotate View> menu item). The left button can be used to pick topography, the middle button will rotate the model when clicked and dragged, and the right button will change the vertical exaggeration when clicked and moved up (more exaggeration) or down (less exaggeration).

Sites and Routes

Sites are individual named locations; routes are named sequences of waypoints, such as survey lines or vehicle missions. Both can be read from files, created and edited interactively, and saved (see the site and route file formats below). To edit sites, select the <Mouse->Edit Sites> mouse mode; the middle button adds a site, the left button selects and drags a site, and the right button deletes the selected site. To edit routes, select the

<Mouse->Edit Routes> mouse mode; with no route selected the middle button starts a new route, with a waypoint selected the middle button adds a new waypoint after it, the left button selects and drags waypoints, and the right button deletes the selected waypoint. Route segments are draped on the topography, and the route lengths both in plan view and along the topography are shown in the Route List. Routes can be saved in the MB-System route format or exported in several formats used by survey and vehicle control software.

Navigation and Swath Data

The navigation of swath (or other) data files can be displayed by opening an **MB-System** datalist with <Action->Open Navigation> or <Action->Open Swath Data>. The navigation is plotted for the parts of the files lying within the grid; <Action->Open Swath Data> also gives the outline of the swath coverage. Where the ancillary navigation (".fnv") or bathymetry (".fbt") files made by **mbdatalist** or **mbprocess** exist, they are read instead of the full data files. Using the <Mouse->Select Nav> or <Mouse->Select Nav File> mouse modes, sections of navigation or whole files can be selected and then opened in the editing and analysis programs listed in the <Action> menu.

Profiles

The <View->Profile> item opens a window showing the topographic profile along the current two point pick, the selected route, or the selected navigation, whichever was picked most recently. The profile can be saved with <Action->Save Profile File>.

Regions and Survey Planning

Using the <Mouse->Pick Area> mouse mode, the left button drags out a rectangular region with north-south and east-west boundaries, and the middle button drags out the centerline of an arbitrarily oriented rectangular area whose width is changed with the right button. A region can be opened as a new display with <Action->Open Region as New View>. An area can be used to generate a survey route with <Action->Generate Survey Route from Area>.

GIS Polylines, Polygons, and Points

MBgrdviz can display GIS polylines (lines with one or more parts), polygons (one or more closed rings, each an outer boundary or a hole), and points, such as survey areas, cable routes, or mooring locations. These are read from MB-System polyline, polygon, and point files (see below) using the <Action->Open Polyline File>, <Action->Open Polygon File>, and <Action->Open Point File> menu items, or imported from ESRI shapefiles using <Action->Import Shapefile>. A shapefile import reads polylines, polygons, or points according to the geometry of the shapefile, transforms coordinates in any projection given by the shapefile's ".prj" file to longitude and latitude, and keeps only the features that overlap the grid displayed in the window from which the import was started; features crossing the edge of the grid are kept whole. Feature names are taken from a NAME, LABEL, or TITLE attribute, else from an ID attribute (as "layer id"), else from the first text attribute. Features without COLOR and SIZE attributes are displayed blue-green (polylines), yellow (polygons), or green (points).

Each file read or imported becomes a layer. The polylines and the polygon outlines are draped on the topography; parts lying off the grid are drawn at the top of the grid's depth range. Points are drawn as triangles. The GIS data are shared by all **MBview** windows like sites and routes, and are displayed or hidden in each window with the <View->Polylines>, <View->Polygons>, and <View->Points> toggles. The <Controls->Polyline List>, <Controls->Polygon List>, and <Controls->Point List> windows list the layers and features; there features and whole layers can be recolored, shown or hidden, and deleted. Polylines can be converted to routes (which can then be edited and saved as route files), and routes can be converted to polylines. The GIS data are not otherwise edited. All features of each type can be saved to MB-System files or exported as shapefiles.

MB-SYSTEM AUTHORSHIP

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OPTIONS

−−help −H

This "help" flag cause the program to print out a description of its operation and then exit immediately.

−−grid−file=grdfile −Igrdfile

Sets a GMT grid file to be read and displayed at startup. This option is usually used only when **MBgrdviz** is started automatically from some other process.

−−overlay−file=grdfile −Jgrdfile

Sets a GMT grid file to be read and displayed as an overlay at startup. This option has no effect unless the **−I** option is also used to specify a primary grid file.

−−test −T

Sets a test grid to be calculated and displayed at startup.This allows the program to be tested even when a grid file is unavailable.

−−verbose −V

Normally, **MBgrdviz** outputs limited information to the stderr stream. If the **−V** flag is given, then **MBgrdviz** works in a "verbose" mode and outputs the program version being used, all error status messages, and a large amount of other information including all of the beams flagged or zeroed.

INTERACTIVE CONTROLS: MBgrdviz Startup Window

[MBgrdviz]:<File->Open Primary Grid>

This menu button pops up a file browser that allows the user to select an input **GMT** grid file. If the OK button is clicked, **MBgrdviz** will attempt to read a topography grid from the specified file. If successful, a new **MBview** window will appear displaying the topography. This item is active until all ten **MBview** windows are in use.

[MBgrdviz]:<File->Open Site>

This menu button pops up a file browser that allows the user to select an input site file (see the site file format below). A site file specifies one or more individual locations that can be displayed and interactively moved. Sites are shared by all **MBview** windows. This item, like the other file items below, is active once an **MBview** window is open, and acts on the first **MBview** window; the <Action> menu of each **MBview** window has the same items acting on that window.

[MBgrdviz]:<File->Open Route>

This menu button pops up a file browser that allows the user to select an input route file (see the route file format below).

[MBgrdviz]:<File->Open Navigation>

This menu button pops up a file browser that allows the user to select an **MB-System** datalist of data files whose navigation is to be displayed (see <Action->Open Navigation>).

[MBgrdviz]:<File->Open Swath>

This menu button pops up a file browser that allows the user to select an **MB-System** datalist of swath data files whose navigation and swath coverage are to be displayed (see <Action->Open Swath

Data>).

[MBgrdviz]:<File->Open Vector>
This menu button pops up a file browser that allows the user to select a vector file (see the vector file format below).

[MBgrdviz]:<File->Open Polyline>

[MBgrdviz]:<File->Open Polygon>

[MBgrdviz]:<File->Open Point>
These menu buttons pop up a file browser that allows the user to select an MB-System polyline (".pln"), polygon (".plg"), or point (".pts") file to be read and displayed in the first **MBview** window, as with the corresponding items of the **MBview** <Action> menu.

[MBgrdviz]:<File->Save Site>
This menu button pops up a file browser that allows the user to specify an output site file to which all sites are written. It is active when sites exist.

[MBgrdviz]:<File->Save Route>
This menu button pops up a file browser that allows the user to specify an output route file to which the selected routes, or all routes if none are selected, are written. It is active when routes exist.

[MBgrdviz]:<File->Quit>
This button causes the program to exit (mostly) gracefully.

[MBgrdviz]:<Realtime->Setup>

[MBgrdviz]:<Realtime->Start>

[MBgrdviz]:<Realtime->Stop>

[MBgrdviz]:<Realtime->Pause>

[MBgrdviz]:<Realtime->Resume>
The **Realtime** menu and its setup dialog are intended for the display of realtime navigation, which is not yet implemented; the menu is inactive.

[MBgrdviz]:<Help->About>
This button causes the program to bring up a dialog showing the program's name, version, and authors.

INTERACTIVE CONTROLS: MBview Main Window

[MBview]:<View->Map Display>

[MBview]:<View->3D Display>
These items select a 2D map view or a 3D perspective view. The map display is not available with the spheroid display projection, which is always shown in 3D.

[MBview]:<View->Topography>

[MBview]:<View->Topography Slope>

[MBview]:<View->Overlay>
These items select the grid that is colored: the primary topography, the magnitude of the topographic slope, or the overlay grid (available once an overlay grid has been read with <Action->Open Overlay Grid>). Each has its own colortable, color bounds, histogram equalization, and shading settings, which are restored when it is selected again.

[MBview]:<View->Histogram Equalization>
This toggle colors the current grid using histogram equalization instead of a linear stretch between the color bounds.

[MBview]:<View->No Shading>

[MBview]:<View->Shading by Illumination>

[MBview]:<View->Shading by Slope>

[MBview]:<View->Shading by Overlay>
These items select the shading of the current grid: none, illumination by a light source, the slope magnitude, or the values of the overlay grid (available once an overlay grid has been read). The shading parameters are set in the <Controls->Shading> dialog or with the <Mouse->Shading> mouse mode.

[MBview]:<View->Topography Contours>
This toggle displays topographic contours at the interval set in the <Controls->Colors & Contours> dialog.

[MBview]:<View->Sites>

[MBview]:<View->Routes>
These toggles display or hide the sites and routes. They are active once site or route display is enabled. Hiding sites or routes also leaves the corresponding editing mouse mode.

[MBview]:<View->Navigation>

[MBview]:<View->Navigation Swath Bounds>

[MBview]:<View->Draped Navigation>
These toggles display the navigation tracks, the outlines of the swath coverage, and the navigation draped on the topography. They are active once navigation has been read.

[MBview]:<View->Vector>
This toggle displays the vector data, shown as colored balls at each point; it is active once vector data have been read.

[MBview]:<View->Polylines>

[MBview]:<View->Polygons>

[MBview]:<View->Points>
These toggles display or hide the GIS polylines, polygons, and points in this window. They are active once data of the type have been read or imported, and are turned on when data are first read into the window.

[MBview]:<View->Haxby Colortable>

[MBview]:<View->Bright Rainbow Colortable>

[MBview]:<View->Muted Rainbow Colortable>

[MBview]:<View->Grayscale Colortable>

[MBview]:<View->Flat Gray Colortable>
These items select the colortable for the current grid. The flat gray colortable is a single shade of gray, useful with shading.

[MBview]:<View->Sealevel1 Colortable>

[MBview]:<View->Sealevel2 Colortable>
These colortables color values below zero (sea level) with the Haxby colors and values above zero with land colors: pale yellow to brown for Sealevel1, and pale yellow through browns to green for Sealevel2.

[MBview]:<View->Profile>
This item opens or closes the profile window (see the Profile window below).

[MBview]:<Controls->Colors & Contours>
This item opens the Colors and Contours dialog (see below).

[MBview]:<Controls->2D Controls>
This item opens the 2D Parameters dialog (see below).

[MBview]:<Controls->3D Controls>
This item opens the 3D Parameters dialog (see below).

[MBview]:<Controls->Shading>
This item opens the Shading Parameters dialog (see below).

[MBview]:<Controls->Resolution>
This item opens the Rendering Resolution dialog (see below).

[MBview]:<Controls->Projections>
This item opens the Projections dialog (see below).

[MBview]:<Controls->Site List>

[MBview]:<Controls->Route List>

[MBview]:<Controls->Navigation List>
These items open the windows listing the sites, routes, and navigation files (see below). The lists are shared by all **MBview** windows.

[MBview]:<Controls->Polyline List>

[MBview]:<Controls->Polygon List>

[MBview]:<Controls->Point List>
These menu buttons bring up windows listing the GIS polylines, polygons, or points (see the Polyline List, Polygon List, and Point List windows below). Like <Controls->Navigation List>, they are always active, even before any data of the type have been read or imported.

[MBview]:<Mouse->Pan & Zoom>
This mouse mode (the default) uses the left button to pick, the middle button to pan, and the right button to zoom (see MOUSE ACTIONS below).

[MBview]:<Mouse->Rotate Model>
This mouse mode, available in the 3D display, uses the middle button to rotate the topography model and the right button to change the vertical exaggeration.

[MBview]:<Mouse->Rotate View>
This mouse mode, available in the 3D display, uses the middle button to rotate the viewpoint and the right button to zoom.

[MBview]:<Mouse->Shading>
This mouse mode uses the middle button to move the light source (for shading by illumination) and the right button to change the shading magnitude.

[MBview]:<Mouse->Pick Area>
This mouse mode uses the left button to define a region, the middle button to define an area, and the right button to change the width of the area.

[MBview]:<Mouse->Edit Sites>
This mouse mode, active once site display is enabled, is used to select, move, add, and delete sites. It turns on the display of sites.

[MBview]:<Mouse->Edit Routes>
This mouse mode, active once route display is enabled, is used to select routes and to add, move, and delete waypoints. It turns on the display of routes.

[MBview]:<Mouse->Select Nav>

[MBview]:<Mouse->Select Nav File>
These mouse modes, active once navigation has been read, select sections of navigation or whole navigation files with the middle button and deselect them with the right button.

[MBview]:<Action->About MBview...>
This item brings up a dialog showing information about **MBview**.

[MBview]:<Action->Open Overlay Grid>
This item pops up a file browser that allows the user to select a **GMT** grid to be displayed as an overlay on the topography, either colored in place of the topography (<View->Overlay>) or used for shading (<View->Shading by Overlay>).

[MBview]:<Action->Open Site File>

[MBview]:<Action->Open Route File>
These items pop up a file browser that allows the user to select a site file (".ste") or route file (".rte") to be read and displayed.

[MBview]:<Action->Open Navigation>

[MBview]:<Action->Open Swath Data>
These items pop up a file browser that allows the user to select an **MB-System** datalist (".mb-1"). The navigation of each data file in the datalist, within the bounds of the grid, is read and displayed as a track named for the file; processed files are used where the datalist calls for them, and ASCII xyz files are skipped. <Action->Open Navigation> reads the ".fnv" navigation files where they exist; <Action->Open Swath Data> reads the ".fbt" bathymetry files where they exist and also gives the outline of the swath coverage. The navigation is decimated according to the grid spacing.

[MBview]:<Action->Open Vector File>
This item pops up a file browser that allows the user to select a vector file (see the vector file format below), which is displayed as colored balls along its tracks.

[MBview]:<Action->Open Polyline File>

[MBview]:<Action->Open Polygon File>

[MBview]:<Action->Open Point File>
These menu buttons pop up a file browser that allows the user to select an MB-System polyline (".pln"), polygon (".plg"), or point (".pts") file. All of the features in the file are read and displayed, whether or not they overlap the grid, with each layer of the file becoming a layer in **MBgrdviz**.

[MBview]:<Action->Import Shapefile>
This menu button pops up a file browser that allows the user to select an ESRI shapefile (".shp"). The polylines, polygons, or points of the shapefile that overlap the grid displayed in this window are imported and displayed as a new layer; the number imported and skipped is printed to the terminal.

[MBview]:<Action->Save Polyline File>

[MBview]:<Action->Save Polygon File>

[MBview]:<Action->Save Point File>
These menu buttons pop up a file browser that allows the user to specify an output file to which all of the polylines, polygons, or points, whether shown or hidden, are written in the MB-System format, keeping their layers, names, colors, and sizes.

[MBview]:<Action->Export Polylines as Shapefile>

[MBview]:<Action->Export Polygons as Shapefile>

[MBview]:<Action->Export Points as Shapefile>
These menu buttons pop up a file browser that allows the user to specify an output ESRI shapefile (".shp" is added if missing). All of the polylines, polygons, or points are written in WGS84 longitude

and latitude with NAME, COLOR, and SIZE attributes (and FILLCOLOR for polygons); an existing shapefile of the same name is replaced.

[MBview]:<Action->Save Site File>
This item pops up a file browser that allows the user to specify an output site file to which all sites are written. It is active when sites exist.

[MBview]:<Action->Save Route File>
This item pops up a file browser that allows the user to specify an output route file to which the selected routes, or all routes if none are selected, are written. It is active when routes exist, as are the other route export items below.

[MBview]:<Action->Save Route File Reversed>
This item writes the selected routes, or all routes, as a route file with the points of each route in reverse order.

[MBview]:<Action->Save Risi Script File (variable heading)>

[MBview]:<Action->Save Risi Script File (static heading)>

[MBview]:<Action->Save Risi 2 Script File (variable heading)>

[MBview]:<Action->Save Risi 2 Script File (static heading)>
These items write the selected routes, or all routes, as mission scripts for the Risi vehicle controller, with the waypoints given in meters relative to the first waypoint. The Risi 2 scripts give each leg as a relative turn ("heading rel") and a move ("move bearing range, bearing", with the bearing relative to the vehicle heading), assuming that the vehicle starts at the first waypoint with heading 0. With variable heading the vehicle turns to face along each leg; with static heading it turns once to the bearing of the first leg, keeps that heading on all legs, and turns back to heading 0 at the end.

[MBview]:<Action->Save Route as Degrees + Decimal Minutes File>
This item writes the waypoints of all routes as latitude and longitude in degrees and decimal minutes.

[MBview]:<Action->Save Route as Hypack LNW File>
This item writes all routes as a Hypack line (".lnw") file, with the waypoints in UTM eastings and northings.

[MBview]:<Action->Save Route as Greensea YML File>
This item writes the selected route (exactly one route must be selected, or exist) as a Greensea mission YAML file.

[MBview]:<Action->Save Route as TECDIS LST File>
This item writes the waypoints of all routes as a TECDIS route list file.

[MBview]:<Action->Save Route as Kongsberg DP waypoint File>
This item writes all routes as a Kongsberg dynamic positioning waypoint file.

[MBview]:<Action->Save Route as SIS ASCIIPlan line File>

[MBview]:<Action->Save Route as SIS ASCIIPlan route File>
These items write all routes as Kongsberg SIS ASCII survey plan files, either as separate lines between consecutive waypoints or as one line per route.

[MBview]:<Action->Save Profile File>
This item writes the current profile (see the profile file format below). It is active when there is a two point pick, a selected route, or selected navigation.

[MBview]:<Action->Open Selected Nav in MBedit>

[MBview]:<Action->Open Selected Nav in MBnavedit>

[MBview]:<Action->Open Selected Nav in MBvelocitytool>
These items start **MBedit**, **MBnavedit**, or **MBvelocitytool** on the data files containing the selected navigation. They are active when navigation has been selected.

[MBview]:<Action->Open Selected Nav in MBeditviz>
This item writes a temporary datalist of the data files containing the selected navigation and starts **MBeditviz** on it.

[MBview]:<Action->Open Region as New View>
This item opens a new **MBview** window displaying the part of the topography (and of any overlay grid) within the current region, with the same display settings. It is active when a region has been picked.

[MBview]:<Action->Generate Survey Route from Area>
This item opens the survey route dialog (see below), which generates a survey route covering the current area. It is active when an area has been picked.

[MBview]:<Dismiss>
This button closes the **MBview** window.

[MBview]:<Full>
This button redraws the display at full resolution.

[MBview]:<Reset>
This button resets the view to the default: map or plan view of the whole grid with no rotation and a vertical exaggeration of 1. Typing "r" in the display does the same.

[MBview]:<Clear>
This button clears the current picks, region, area, profile, and the selections of navigation, sites, and routes.

[MBview]:<Pan & Zoom>

[MBview]:<Rotate Model>

[MBview]:<Rotate View>

[MBview]:<Shading>

[MBview]:<Pick Area>

[MBview]:<Edit Sites>

[MBview]:<Edit Routes>

[MBview]:<Pick Nav>

[MBview]:<Pick Nav File>
When the **MBview** window is taller than 700 pixels, these buttons on its left side select the same mouse modes as the items of the <Mouse> menu.

INTERACTIVE CONTROLS: MBview Colors and Contours Window

Changes take effect when the **Apply** button is pressed.

[Colors and Contours]:{Data Color Bounds}"Minimum"

[Colors and Contours]:{Data Color Bounds}"Maximum"

[Colors and Contours]:{Data Color Bounds}<<Cold-to-Hot><Hot-to-Cold>>
These set the range of topography values spanned by the colortable, and whether the colortable runs from cold to hot or from hot to cold colors.

[Colors and Contours]:"Data Contour Interval (meters)"
This sets the interval of the topographic contours.

[Colors and Contours]:{Data Slope Color Bounds}"Minimum"

[Colors and Contours]:{Data Slope Color Bounds}"Maximum"

[Colors and Contours]:{Data Slope Color Bounds}<<Cold-to-Hot><Hot-to-Cold>>
These set the color bounds and direction for the topographic slope display.

[Colors and Contours]:{Overlay Color Bounds}"Minimum"

[Colors and Contours]:{Overlay Color Bounds}"Maximum"

[Colors and Contours]:{Overlay Color Bounds}<<Cold-to-Hot><Hot-to-Cold>>
These set the color bounds and direction for the overlay grid display.

[Colors and Contours]:<Apply>
This button applies the values and redraws the display.

[Colors and Contours]:<Dismiss>
This button closes the dialog.

INTERACTIVE CONTROLS: MBview 2D Parameters Window

Changes take effect when the **Apply** button is pressed.

[2D Parameters]:"Zoom"
This sets the size of the 2D view; larger values zoom in.

[2D Parameters]:{View Offset}"X"

[2D Parameters]:{View Offset}"Y"
These set the horizontal offset of the 2D view.

[2D Parameters]:<Apply>

[2D Parameters]:<Dismiss>
These buttons apply the values and close the dialog.

INTERACTIVE CONTROLS: MBview 3D Parameters Window

Changes take effect when the **Apply** button is pressed.

[3D Parameters]:{Model Orientation & Zoom}"Azimuth (degrees)"

[3D Parameters]:{Model Orientation & Zoom}"Elevation (degrees)"

[3D Parameters]:{Model Orientation & Zoom}"Zoom"
These set the orientation and zoom of the topography model.

[3D Parameters]:{View Orientation & Zoom}"Azimuth (degrees)"

[3D Parameters]:{View Orientation & Zoom}"Elevation (degrees)"

[3D Parameters]:{View Orientation & Zoom}"Zoom"
These set the orientation and zoom of the viewpoint.

[3D Parameters]:{Pan}"X"

[3D Parameters]:{Pan}"Y"
These set the horizontal offset of the 3D view.

[3D Parameters]:"Vertical exaggeration"
This sets the vertical exaggeration of the 3D display.

[3D Parameters]:<Apply>

[3D Parameters]:<Dismiss>
These buttons apply the values and close the dialog.

INTERACTIVE CONTROLS: MBview Shading Parameters Window

Changes take effect when the **Apply** button is pressed.

[Shading Parameters]:{Shading by Illumination}"Amplitude"

[Shading Parameters]:{Shading by Illumination}"Azimuth (degrees)"

[Shading Parameters]:{Shading by Illumination}"Elevation (degrees)"
These set the strength of the illumination shading and the direction of the light source.

[Shading Parameters]:{Shading by Slope}"Amplitude"
This sets the strength of the slope shading.

[Shading Parameters]:{Shading by Overlay}"Amplitude"

[Shading Parameters]:{Shading by Overlay}"Center"

[Shading Parameters]:{Shading by Overlay}<<Cold-to-Hot><Hot-to-Cold>>
These set the strength of the shading by the overlay grid, the overlay value giving no shading, and the direction of the shading.

[Shading Parameters]:<Apply>

[Shading Parameters]:<Dismiss>
These buttons apply the values and close the dialog.

INTERACTIVE CONTROLS: MBview Rendering Resolution Window

The sliders take effect when released. While the display is being moved, it is drawn at low resolution, then at medium resolution when the mouse button is released; <Full> draws it at full resolution.

[Rendering Resolution]:{Grid Rendering Resolution}==Medium Resolution Dimension==

[Rendering Resolution]:{Grid Rendering Resolution}==Low Resolution Dimension==
These set the largest number of grid cells drawn along each side of the grid at medium and low resolution (rounded to multiples of 25; the low resolution dimension is no larger than the medium one).

[Rendering Resolution]:{Nav Rendering Decimation}==Medium Resolution Decimation==

[Rendering Resolution]:{Nav Rendering Decimation}==Low Resolution Decimation==
These set the decimation (every nth point) of the navigation and vector data drawn at medium and low resolution. Like routes, GIS polylines and polygons are always drawn at full resolution.

[Rendering Resolution]:<Dismiss>
This button closes the dialog.

INTERACTIVE CONTROLS: MBview Projections Window

The toggles take effect immediately.

[Projections]:{Display Projection}<<Geographic><UTM><Spheroid>>
These select the projection of the display: geographic coordinates, the UTM zone of the grid center, or a spheroidal (globe) view, which is always displayed in 3D.

[Projections]:{Annotation Style}<<Degrees + Minutes><Decimal Degrees>>
These select how longitudes and latitudes are shown in the pick information and in the lists.

[Projections]:"Primary Grid Projection"
This shows the projection of the primary grid.

[Projections]:<Dismiss>
This button closes the dialog.

INTERACTIVE CONTROLS: MBview Site List Window

[Site List]:|Site List|

This list shows each site with its number, longitude, latitude, depth, color, size, and name. Selecting a site highlights it in the displays; clicking the selected site a second time changes its color to the next color (skipping red, which marks the selection). Several sites can be selected with the shift and control keys.

[Site List]:<Delete Selected Sites>

This button deletes the selected sites.

[Site List]:<Dismiss>

This button closes the list window.

INTERACTIVE CONTROLS: MBview Route List Window

[Route List]:|Route List|

This list shows each route, with its number, editability, number of waypoints, length in plan view and along the topography, color, and name, followed by its waypoints with their longitude, latitude, depth, distances along the route, and waypoint type. Selecting a route or waypoint highlights it in the displays. Clicking a selected route a second time changes its color; clicking a selected waypoint a second time changes its waypoint type (simple, transit, or start or end of a survey line of types 1 to 5).

[Route List]:<Delete Selected Routes>

This button deletes the selected routes and waypoints.

[Route List]:<Convert Selected to Polylines>

This button, present only when GIS polylines are displayed, converts each selected route (selecting a route or any of its waypoints selects the route) to a polyline through its waypoints, with the route's color, size, and name and the topography at the waypoints. The polylines are placed in a layer named "Converted from routes"; the routes are kept.

[Route List]:<Dismiss>

This button closes the list window.

INTERACTIVE CONTROLS: MBview Navigation List Window

[Navigation List]:|Navigation List|

This list shows each navigation file with its number, number of points, color, size, and name. Selecting files selects all of their navigation, as with the <Mouse->Select Nav File> mouse mode, and the profile follows the selected navigation.

[Navigation List]:<Delete Selected Navigation>

This button deletes the selected navigation files from the display.

[Navigation List]:<Dismiss>

This button closes the list window.

INTERACTIVE CONTROLS: MBview Profile Window

[Profile]:|Profile|

The profile window shows the topography along the current two point pick, selected route, or selected navigation, with the source, length, and depth range above the plot. Segments steeper than the slope threshold are drawn in red, and the boundaries between route or pick segments in green.

[Profile]:==Display Width Multiple==

This sets the width of the plot as a multiple (1 to 5) of the window width.

[Profile]:==Slope Threshold==
This sets the slope (from 0.01 to 2.00) above which segments are drawn in red.

[Profile]:==Vertical exaggeration==
This sets the vertical exaggeration of the profile.

[Profile]:<Dismiss>
This button closes the profile window.

INTERACTIVE CONTROLS: Generate Survey Route Window

This dialog, opened with <Action->Generate Survey Route from Area>, generates a route of parallel survey lines covering the current area, run back and forth along the length of the area, optionally with crosslines. Each line end is a waypoint of type "start line". The route is generated when **Generate Route** is pressed; pressing it again with the same route name replaces the route. The "Current Area Info" shows the area dimensions and the resulting route.

[Survey Route]:"Route Name"

[Survey Route]:"Route Color"
These set the name (default "Survey") and color of the route.

[Survey Route]:"Line Spacing Mode"
This selects uniform line spacing or line spacing varying with the swath width of the sonar.

[Survey Route]:"Line Spacing (m)"
This sets the spacing of the lines in the uniform mode.

[Survey Route]:"Platform Type"

[Survey Route]:"Swath Width (degrees)"

[Survey Route]:"Sonar Depth (m)"

[Survey Route]:"Sonar Altitude (m)"
In the variable mode these set whether the sonar is on a surface vessel, a submerged vehicle at constant altitude, or a submerged vehicle at constant depth, the sonar swath width, and the sonar depth or altitude. For constant altitude the line spacing follows from the altitude and swath width; otherwise each line spacing is calculated from the shallowest topography along the previous line.

[Survey Route]:"Start Corner"
This sets the corner of the area (SW, SE, NW, or NE) where the survey starts.

[Survey Route]:"Interleaving"
This sets the number of passes over the area; with values greater than 1 the lines are run in interleaved passes.

[Survey Route]:"Cross Lines"

[Survey Route]:"Crosslines Before/After Survey"
These set the number of crosslines across the survey lines and whether they are run before or after the survey lines.

[Survey Route]:<Generate Route>

[Survey Route]:<Dismiss>
These buttons generate the route and close the dialog.

INTERACTIVE CONTROLS: MBview Polyline, Polygon, and Point List Windows

[Polyline List]:|Polyline List|

[Polygon List]:|Polygon List|

[Point List]:|Point List|

These lists show each layer of the type, followed by its features. A layer item shows whether the layer is shown or hidden, the number of features, and the layer name; a feature item shows whether it is shown or hidden, its color, its number of points and parts or rings (or for points the position and z value), and its name. Selecting a single feature highlights it in red in the displays. Clicking an item a second time changes its color to the next color (skipping red); clicking a layer item a second time changes the color of all of its features. Several items can be selected with the shift and control keys.

[Polyline List]:<Show/Hide Selected>

[Polygon List]:<Show/Hide Selected>

[Point List]:<Show/Hide Selected>

This button hides the selected features and layers if they are shown, or shows them if they are hidden. Hidden features are still saved and exported.

[Polyline List]:<Delete Selected>

[Polygon List]:<Delete Selected>

[Point List]:<Delete Selected>

This button deletes the selected features; selecting a layer selects all of its features.

[Polyline List]:<Convert Selected to Routes>

This button converts each part of each selected polyline (or of all polylines of a selected layer) to an editable route with a waypoint at each vertex, keeping the polyline's color, size, and name (with " part n" added for polylines with several parts). Vertices that cannot be placed in the display are not used, and their number is printed to the terminal. The polylines are kept.

[Polyline List]:<Dismiss>

[Polygon List]:<Dismiss>

[Point List]:<Dismiss>

This button closes the list window.

MOUSE ACTIONS

The actions of the mouse buttons in the display depend on the mouse mode, set with the <Mouse> menu; the current actions are listed in the "Mouse Mode" box on the left side of the window. On release of a button the display is redrawn at medium resolution.

Left Mouse Button

In the Pan & Zoom, Rotate Model, Rotate View, Shading, Select Nav, and Select Nav File modes, clicking picks a point on the topography, marked with a red X and described in the "Pick Info" box; dragging picks two points and reports the range and bearing between them, and gives the profile between them. In the Pick Area mode, dragging defines a region with north-south and east-west boundaries; dragging from near a corner of an existing region moves that corner. In the Edit Sites mode, clicking selects the nearest site and dragging moves it. In the Edit Routes mode, clicking selects the nearest waypoint (and its route) and dragging moves it; clicking off the grid clears the selection.

Middle Mouse Button

In the Pan & Zoom mode, dragging pans the view. In the Rotate Model and Rotate View modes, dragging rotates the model or the viewpoint in azimuth (left and right) and elevation (up and down). In the Shading mode with shading by illumination, dragging moves the light source. In the Pick Area mode, dragging defines the centerline of an area; dragging from near an end of an existing area moves that end. In the Edit Sites mode, clicking adds a new site after the selected site and dragging moves it. In the Edit Routes mode, clicking with no route selected starts a new route; with a waypoint selected, clicking adds a new waypoint after it (after the last waypoint to extend a route, or between

waypoints to insert one), and dragging moves the new waypoint. In the Select Nav mode, dragging selects the navigation between the two picked points; in the Select Nav File mode, it selects all of the navigation files from the first to the second picked file.

Right Mouse Button

In the Pan & Zoom and Rotate View modes, dragging up zooms in and dragging down zooms out. In the Rotate Model mode, dragging up or down increases or decreases the vertical exaggeration. In the Shading mode, dragging up or down increases or decreases the shading magnitude. In the Pick Area mode, dragging up or down widens or narrows the area. In the Edit Sites and Edit Routes modes, clicking on the selected site or waypoint deletes it; deleting the last waypoint of a route deletes the route. In the Select Nav and Select Nav File modes, the right button deselects navigation as the middle button selects it.

KEYBOARD ACTIONS

r or R

Typing "r" or "R" in the display resets the view, as with the **Reset** button.

MBGRDVIZ ROUTE FILES

MB-System route files (suffix ".rte") are text files beginning with a header line "## Route File Version 2.00" and further comment lines beginning with "##" that describe the format. Each route begins with a line "> ## STARTROUTE np,color,size,editmode,name", where np is the number of points, color is a color number as for the GIS files (0 = black to 7 = purple), size is the line width, editmode is 1 if the route can be edited, and the name is the rest of the line, and ends with a line "> ## ENDROUTE". Between these lines are the points of the route, one per line, with the comma separated values:
longitude (deg), latitude (deg), topography (m), waypoint type,
bearing (deg), lateral distance (m), distance along topography (m),
slope (m/m)
The waypoint type is 0 for points that only give the topography between waypoints, 1 for a simple waypoint, 2 for a transit waypoint, and 3 to 12 for the start and end of survey lines of types 1 to 5 (3 = start line, 4 = end line, 5 = start line 2, and so on); waypoints are followed by a comment such as "## WAYPOINT STARTLINE". When a route file is read, only the longitude, latitude, and waypoint type of the waypoints (type 1 or greater) are used; the topography points are recalculated from the grid. Lines beginning with ">" end the current route.

Files without the version header are also read: each line gives at least the longitude and latitude of a waypoint, separated by commas or white space, optionally followed by the topography and the waypoint type, and routes are separated by lines beginning with ">". In such files the comment lines "## ROUTENAME name", "## ROUTECOLOR color", "## ROUTESIZE size", and "## ROUTEEDITMODE mode" set the route properties. For example, a file giving a route of two survey lines and a route of one survey line:

```
## ROUTENAME Lines A
## ROUTECOLOR 1
−121.12 34.00 0 3
−121.02 34.06 0 4
−121.02 34.07 0 3
−121.12 34.01 0 4
>
## ROUTENAME Lines B
−121.12 33.98 0 3
−121.02 34.04 0 4
```

The route export formats of the <Action> menu are written for particular survey and vehicle control systems and are not read by **MBgrdviz**.

MBGRDVIZ SITE FILES

MB-System site files (suffix ".ste") are text files beginning with a header line "## Site File Version 2.00" and further comment lines beginning with "#". Each other line gives one site, with the comma separated values:
longitude (deg), latitude (deg), topography (m), color, size, name
where the color is a color number as for the GIS files (0 = black to 7 = purple), and the name is the rest of the line. Only the longitude and latitude are required. Longitudes and latitudes may also be given as degrees and decimal minutes separated by a colon (e.g. −121:04.2). Files with version 1 headers, or without a header and without commas, are read with white space separated values. For example:

```
## Site File Version 2.00
−121.0800000,  34.0200000, −1650.000, 4, 1,M1
−121.0500000,  34.0500000, −2010.000, 7, 1,M2 deep mooring
```

MBGRDVIZ VECTOR FILES

Vector files are text files giving tracks of points with data values to be displayed as colored balls. Each line gives the white space separated longitude (deg), latitude (deg), z (m), and data value of one point; lines beginning with ">" separate tracks, and lines beginning with "#" are comments, except that "## VECTORNAME name" sets the name (a single word) of the track and "## MIN value" and "## MAX value" set the range of data values spanned by the colors (by default the range of the data).

MBGRDVIZ PROFILE FILES

Profile files written with <Action->Save Profile File> are text files with a header of comment lines beginning with "##", including the profile source, length, and depth range, followed by one line per profile point with the white space separated values: distance along the profile (m), topography (m), segment boundary flag, longitude (deg), latitude (deg), distance along the topography (m), bearing (deg), and slope (m/m).

MBGRDVIZ POLYLINE FILES

MB-System polyline files (suffix ".pln") are text files. The file begins with a header line identifying the file type and version, followed by comment lines beginning with "##". A "## Z values: present" or "## Z values: absent" line records whether the z values are meaningful, and lines of the form
LAYER l,layer name
give the names of the layers. Colors are given as integers: 0 = black, 1 = white, 2 = red, 3 = yellow, 4 = green, 5 = blue-green, 6 = blue, and 7 = purple (red is used to highlight selected features). Sizes are line widths in pixels for polylines and polygons, and larger symbols for points with sizes above 1. Positions are longitude and latitude in decimal degrees (WGS84) and z values are in meters, positive up. Each polyline begins with a line "> ## STARTPOLYLINE npart,npoint,color,size,layer,name", each of its parts begins with a line "> ## PART ipart,npoint" followed by the comma separated longitude, latitude, and z values of its points, and the polyline ends with a line "> ## ENDPOLYLINE". The name is the rest of the line and may contain spaces and commas. Because the "##" lines are comments and the ">" lines are segment headers for **GMT**, the files can also be plotted directly with **GMT**. For example:

```
## Polyline File Version 1.00
## Z values: absent
## LAYER 0,survey lines
> ## STARTPOLYLINE 2,5,1,3,0,Line A
> ## PART 0,3
−121.120000000,34.000000000,0.000
−121.070000000,34.030000000,0.000
−121.020000000,34.060000000,0.000
> ## PART 1,2
−121.120000000,34.060000000,0.000
−121.020000000,34.000000000,0.000
> ## ENDPOLYLINE
```

MBGRDVIZ POLYGON FILES

MB-System polygon files (suffix ".plg") have the same form as polyline files, with polygons beginning with a line "> ## STARTPOLYGON nring,npoint,color,size,fillcolor,layer,name" (the fill color is −1 for no fill; filled display is not yet supported), rings beginning with a line "> ## RING iring,npoint,OUTER" for outer boundaries or "> −Ph ## RING iring,npoint,HOLE" for holes (the −Ph marks a hole for **GMT**), and polygons ending with a line "> ## ENDPOLYGON". Each ring is closed, its last point repeating its first point. For example:

```
## Polygon File Version 1.00
## Z values: absent
## LAYER 0,study areas
> ## STARTPOLYGON 2,10,3,2,−1,0,Closure with hole
> ## RING 0,5,OUTER
−122.000000000,36.000000000,0.000
−121.000000000,36.000000000,0.000
−121.000000000,37.000000000,0.000
−122.000000000,37.000000000,0.000
−122.000000000,36.000000000,0.000
> −Ph ## RING 1,5,HOLE
−121.800000000,36.200000000,0.000
−121.800000000,36.800000000,0.000
−121.200000000,36.800000000,0.000
−121.200000000,36.200000000,0.000
−121.800000000,36.200000000,0.000
> ## ENDPOLYGON
```

MBGRDVIZ POINT FILES

MB-System point files (suffix ".pts") have the same header and layer lines as polyline files, followed by one line per point giving the comma separated longitude, latitude, z, color, size, layer, and name. For example:

```
## Point File Version 1.00
## Z values: absent
## LAYER 0,moorings
−121.080000000,34.020000000,0.000,4,2,0,M1
−121.050000000,34.050000000,0.000,7,1,0,M2
```

SHAPEFILES

MBgrdviz imports and exports ESRI shapefiles using the **GDAL** library. Imported shapefiles may contain points, multipoints (each point becomes a separate feature), lines, multilines, polygons, or multipolygons, with or without z values, in any coordinate system given by a ".prj" file (without a ".prj" file the coordinates are taken to be longitude and latitude). Exported shapefiles use WGS84 longitude and latitude, are encoded in UTF-8, and have NAME, COLOR, and SIZE attributes (and FILLCOLOR for polygons), so that they can be imported again with the same names, colors, and sizes. Shapefile attribute text is limited to 254 bytes, so longer names are shortened on export.

SEE ALSO

mbsystem(1), **mbdatalist**(1), **mbedit**(1), **mbeditviz**(1), **mbnavedit**(1), **mbprocess**(1), **mbvelocitytool**(1)

BUGS

The **Realtime** menu is not yet implemented. Polygons are displayed as outlines only; filled polygons are not yet supported.